

Accessibility of convenient stores across Brussels neighborhoods

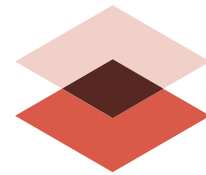
Thomas Ermans & Pierre-François Wilmotte



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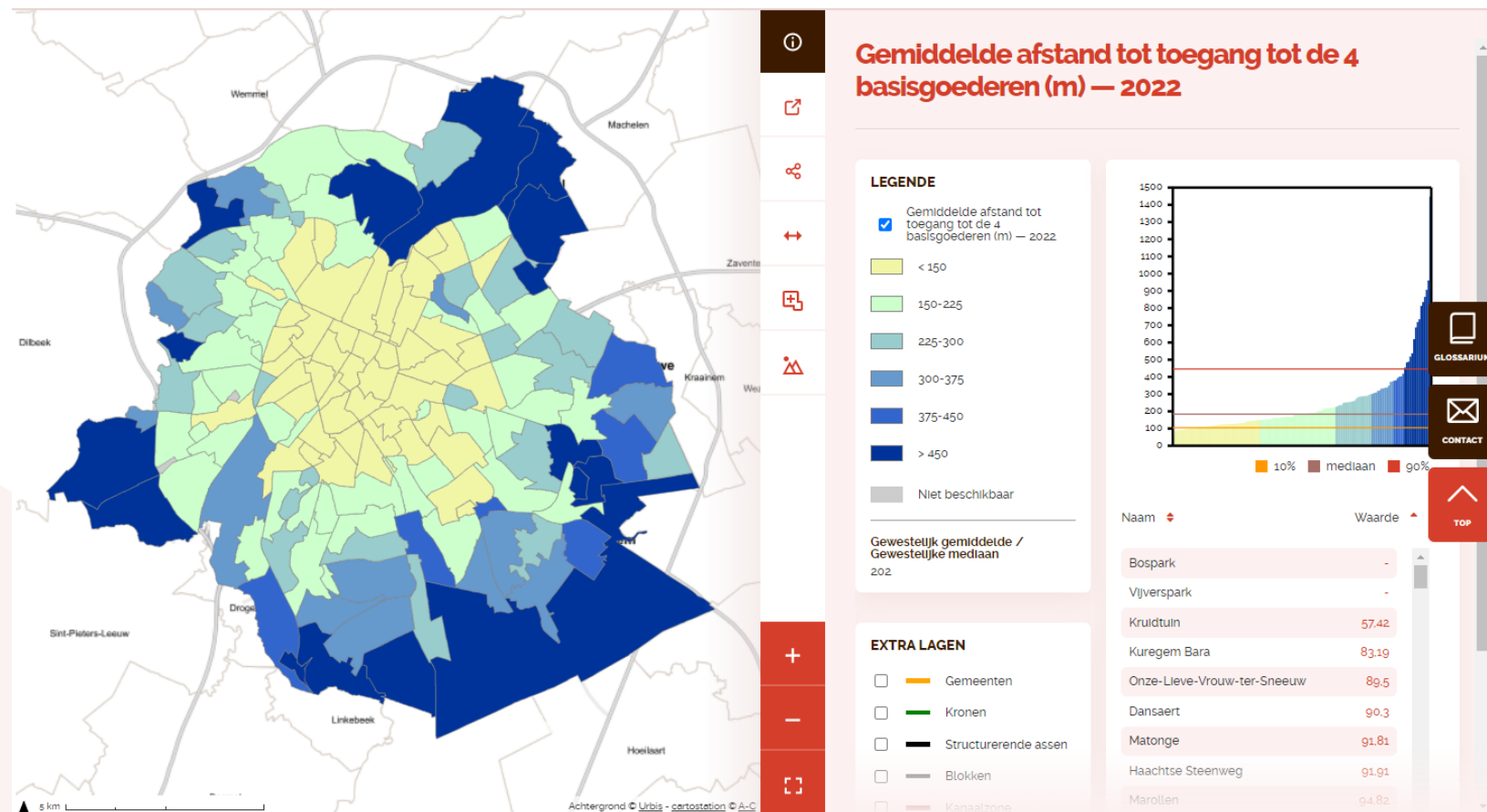
MEASURING LOCAL ACCESSIBILITY THROUGH THE MEAN DISTANCE TO (4) BASIC PRODUCTS

ACCESSIBILITY?

The potential of households to make use of the resources supplied by the (local) environment

SPATIAL accessibility

MEAN DISTANCE TO 4 BASIC PRODUCTS



Winkels: Oligo database (Hub.Brussels)

- › Updated yearly through fieldwork
- › Name, category, location
- › No financial data

Population: National register adress (Statbel)

Bronnen: Hub.Brussels (2022), Statbel (RR) 2022; Berekeningen BISA (2023) via <https://monitoringdesquartiers.brussels/> <https://wijkmonitoring.brussels/>

COMMERCIEËLE VESTIGINGEN IN BRUSSEL EN BELGIË

4 basic products and different regulations

› Bread, meat, general grocery store

- **Freedom to establish businesses** since the 2006 EU Services Directive (end of socio-economic permits) > BUT “Compelling reasons of general interest such as protection of the urban environment, social governance of public health” > urban criteria > **PRAS/GBP**
- **Influenced greatly by economic and market processes, tempered by specific regulations**

› Pharmacies

- Pharmacy: 1 pharmacy/5.000 inhabitants (per municipality) + proximity regulations > **better dispersion over the entire Region**

Soort winkel	Brood	Vlees	Algemene voedingsmiddelen	Farmaceutische producten
Kruidenier			X	
Superette	X	X	X	
Supermarkt	X	X	X	
Hypermarkt	X	X	X	
Dieetvoeding			X	
Delicatessen - Ambachtelijke producten			X	
Kruidenierswinkel gespecialiseerd in buitenlandse producten			X	
Bakkerij	X			
Patisserie	X			
Slagerij - Charcuterie		X		
Groenten en fruit			X	
Apotheek				X
Broodjeszaak	X			

Bron : Wijkmonitoring van BISA

HOW TO CALCULATE THE MEAN DISTANCE TO THE 4 BASIC PRODUCTS?

3 steps

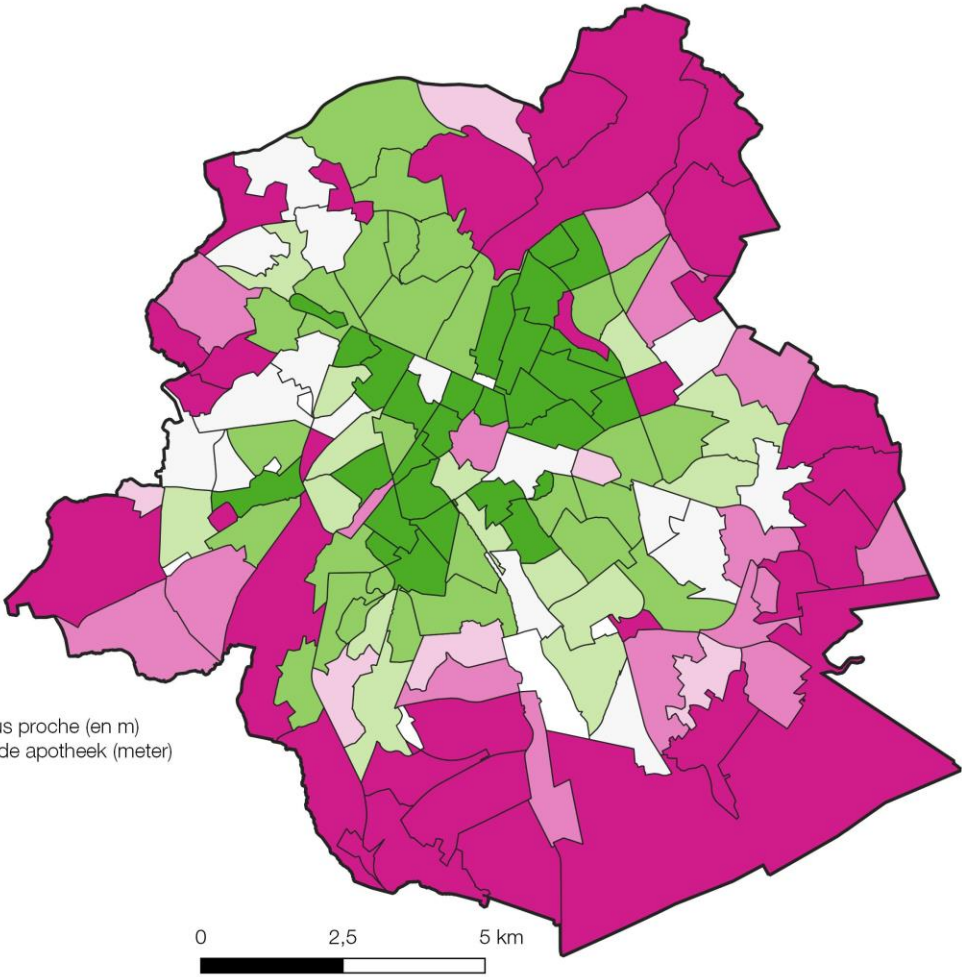
1. The distance as the crow flies between each RN-address and the closest store is computed for each product
2. The mean distance to the 4 closest store is computed for each RN-address
3. Weighted aggregation by population at the desired level

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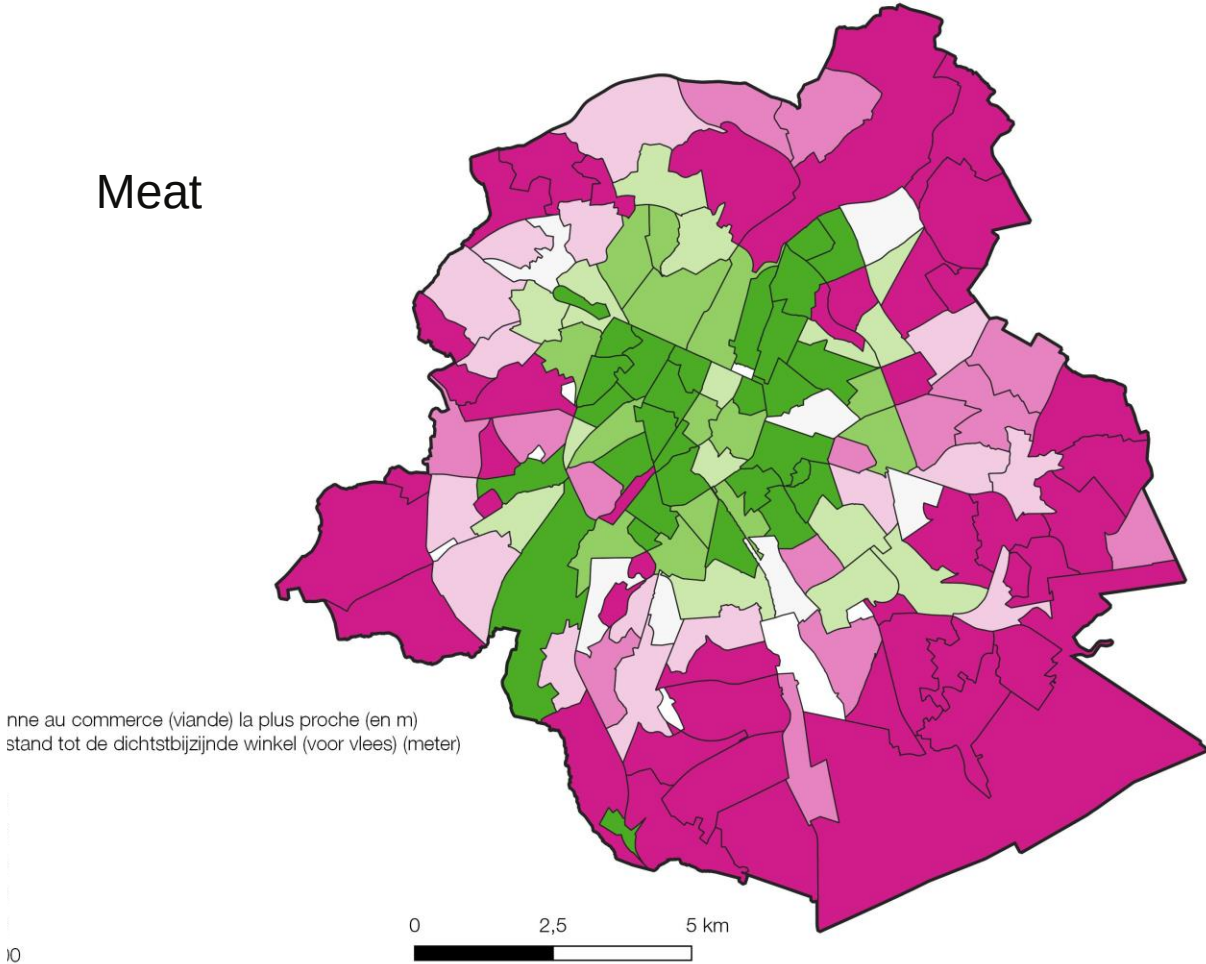
Bron : Wijkmonitoring van BISA

SINGLE-PRODUCT MEAN DISTANCE

Pharmacies



Meat

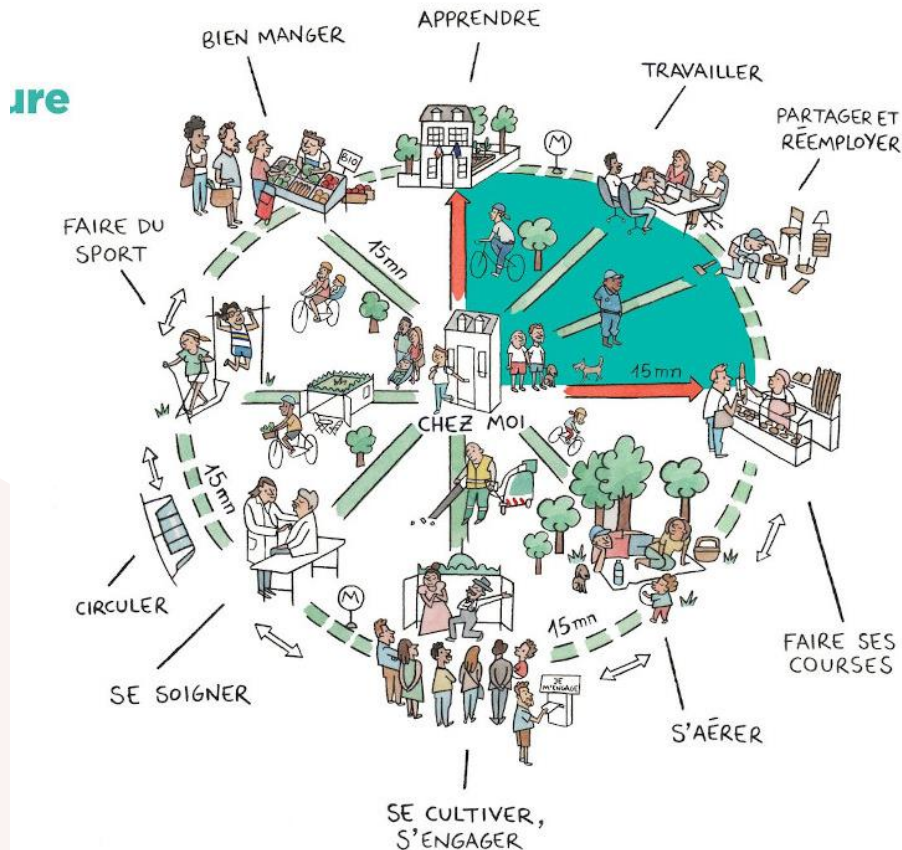


Bronnen : Statbel (RR), 2022; hub.brussels, 2023; Berekeningen BISA, 2025

2 RESEARCH QUESTIONS

- 1. How does the accessibility to convenient stores vary accross neighborhoods in Brussels and is there a link with population distribution?**
- 2. What is the spatial link between accessibility to convenient stores and socioeconomic characteristics of the population?**

CONTEXT



« 15 minutes-city » ↔ Improve spatial accessibility by strengthening the proximity to local stores and shops

Possible insights for public authorities

- Identification of places where accessibility is poor
- Lever to limit nuisances

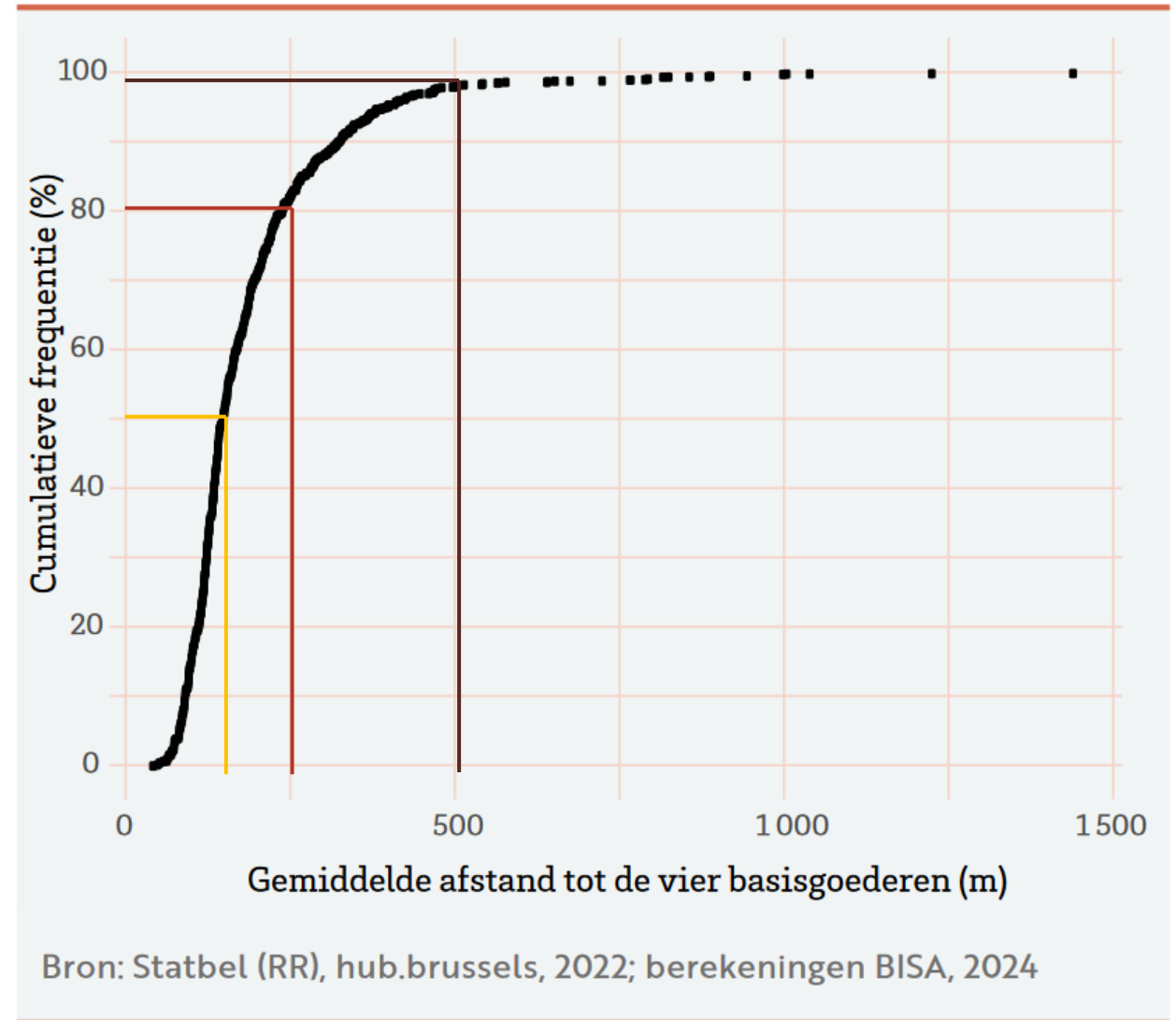
02

ANALYSES



GLOBAL RESULTS

- › 50 % of the Brussels population lives less than 152 meters away from the 4 basic products (and therefore 50% live more than 152 meters away)
- › 81 % of the Brussels population finds the 4 basic products under an average of 250 meters from home
- › 98 % of the Brussels population finds the 4 basic products under an average of 500 meters from home

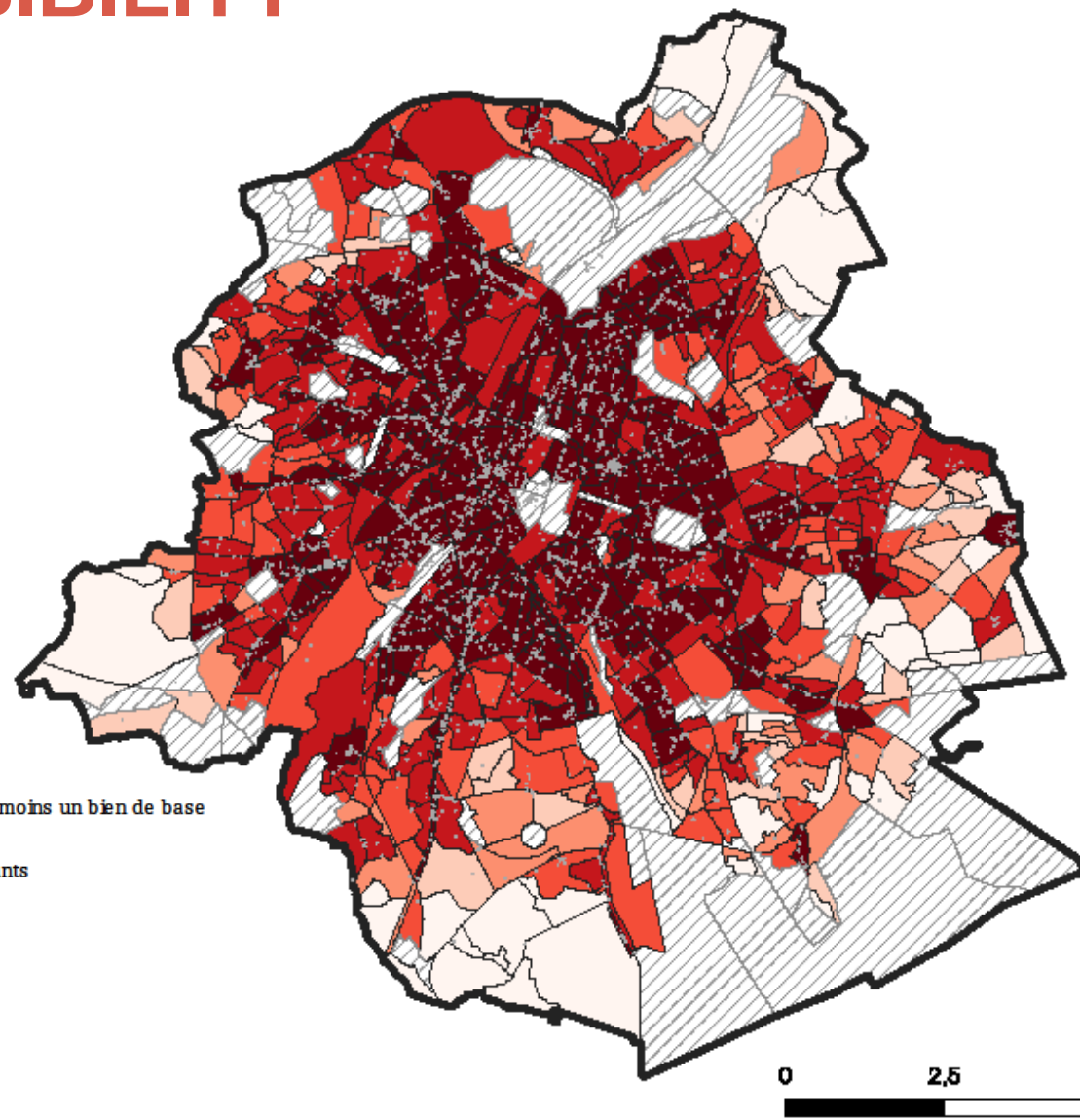


GEOGRAPHY OF THE ACCESSIBILITY TO CONVENIENT STORES

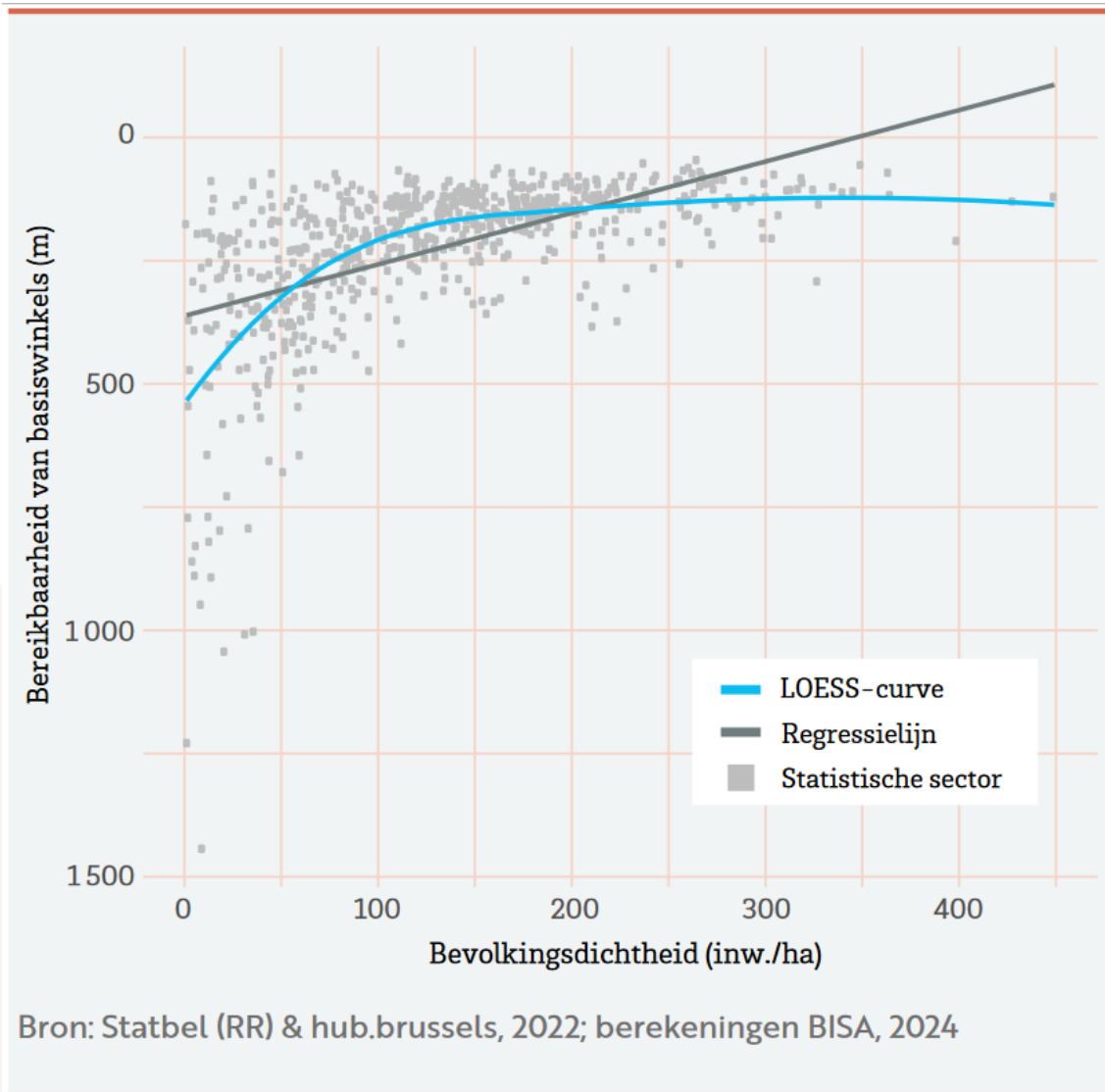
Grondgebied	Gemiddelde (m)	Q1 (m)	Mediaan (m)	Q3 (m)
BHG	189	122	152	220
Vijfhoek	104	89	100	123
Eerste kroon	132	101	128	146
Tweede kroon	231	144	193	270

Observations

- › Center-periphery distribution
- › Effects of local retail supply



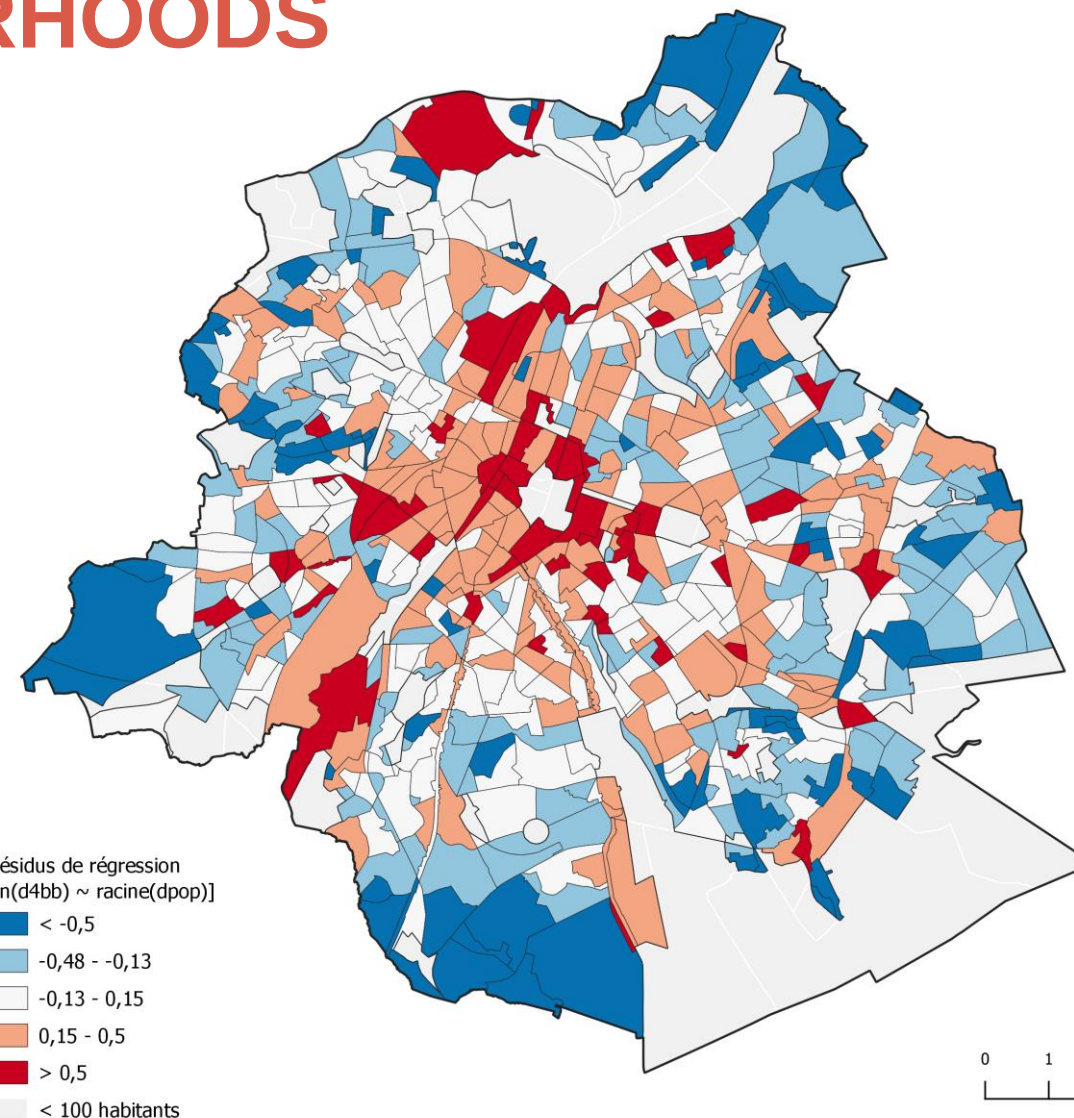
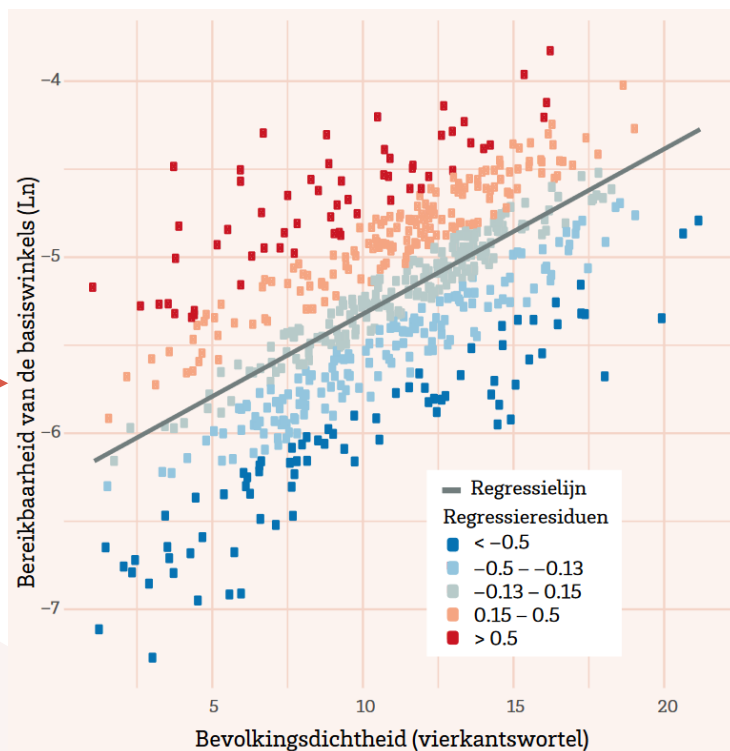
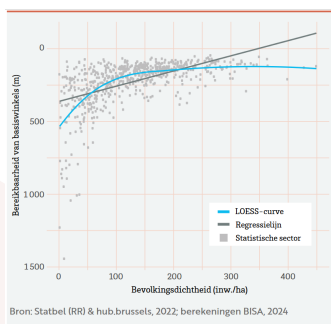
IS THERE A LINK WITH POPULATION DENSITY?



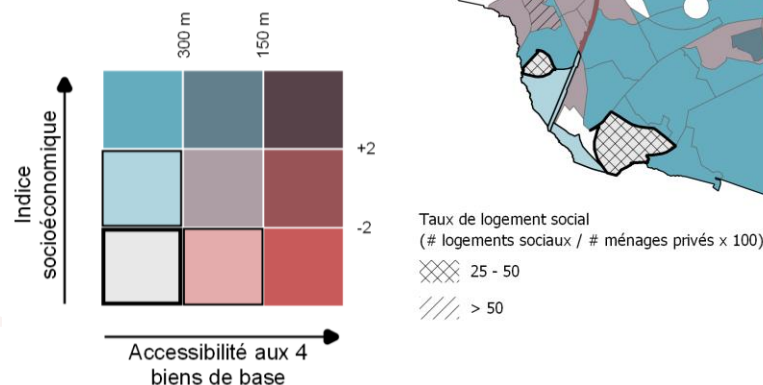
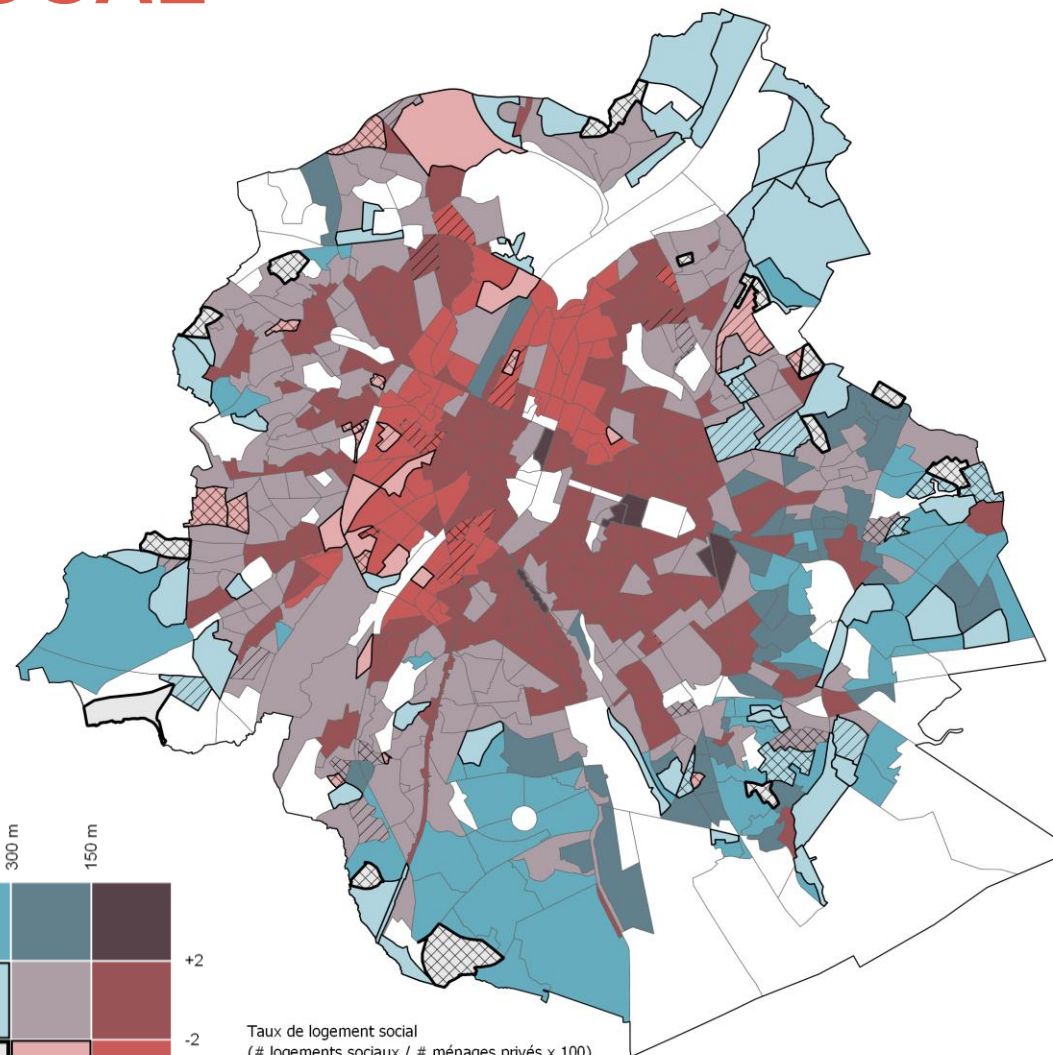
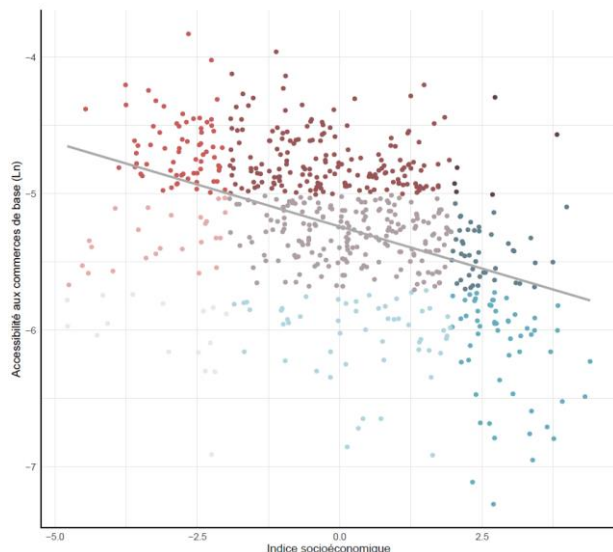
Findings

- › Not linear but positive relation
- › Bigger variability in less dense sectors

BEYOND POPULATION DENSITY, UNDER- AND OVERACCESSIBLE NEIGHBORHOODS



IS THERE A LINK WITH THE LOCAL SOCIOECONOMIC LEVEL?



- **Negative statistical relationship** between local accessibility and socioeconomic index → **positive** w.r.t. social justice
- Neighborhoods with low accessibility **AND** low socioeconomic index (often neighborhoods with social housings) → **negative** w.r.t. social justice

03

CONCLUSIONS EN DISCUSSIONS



CONCLUSIONS EN DISCUSSIONS

- Social justice > second crown
 - Neighborhoods with social housings
 - Poorer households
- Relationship between local accessibility and socioeconomic index can evolve > gentrification of inner-city neighborhoods
- Long-term reconfiguration of the retail infrastructure threatens local accessibility
- Remedial actions
 - Spatial regulations
 - Mobility
 - Access to car mobility



FOCUS N°64

L'accessibilité aux commerces du quotidien à travers les quartiers bruxellois

THOMAS ERMANS & PIERRE-FRANÇOIS WILMOTTE

La proximité entre les lieux de vie et les commerces reste un enjeu important de l'aménagement urbain. Habiter à proximité de commerces de première nécessité est un facteur de localisation résidentielle appréciable. La planification territoriale de cette proximité est un enjeu social et environnemental. À l'aide d'un indicateur quantifiant la distance entre le lieu de résidence et les commerces, ce Focus objective la qualité de la desserte commerciale à travers les quartiers bruxellois.

<https://ibsa.brussels/publications/focus-de-l-ibsa>